

Overnight, or Overhyped?

A Reproducible Audit of the Overnight Return Anomaly
and the Market-Manipulation Hypothesis

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github.com/Guillain-RDCDE/Open-Alpha-Lab

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Abstract

Across world equity markets, close-to-open (“overnight”) returns have historically dwarfed open-to-close (“intraday”) returns. Knuteson [6, 7] reads this as the signature of deliberate manipulation by a large quantitative fund. We take the empirical fact seriously—and confirm it with autocorrelation-robust inference and at the individual-firm level—then audit the inference. We show that (i) the headline magnitudes are inflated by compounding, data artefacts, and above all a *calendar-time* confound: the overnight window spans ~ 28 hours versus the 6.5-hour day, so per hour the edge is $\sim 1.3\times$, not $\sim 4\times$; (ii) what survives is largely a gap-risk premium (beta), sits below realistic trading costs, and has decayed from a 5-year Sharpe near 2 to near 0.5; (iii) the strategy is uninvestable at scale, with square-root market-impact capacity in the single-digit millions of dollars; and (iv) the alleged manipulation mechanism is economically self-defeating—a steelman simulation shows a firm large enough to “move world markets” would lose tens of billions per year running it. A transparent Bayesian analysis returns a low posterior probability of manipulation that is robust across priors. Every number is reproducible from the accompanying open-source toolkit.

1 Introduction

The observation is real and old: U.S. equity index returns accrue overwhelmingly while the market is closed, with the trading day flat to negative [3, 2, 9]. Knuteson extends it across world markets and three decades and advances a striking interpretation: a single large quantitative fund expands its book when the market is illiquid (near the open, moving prices up) and contracts it when liquid, harvesting the overnight drift it helps create [6, 7].

This paper neither dismisses the fact nor accepts the interpretation. We grant the empirical pattern its fairest hearing, then ask what it actually implies. Our contribution is an end-to-end, fully reproducible audit that separates three levels routinely conflated in the popular framing: the *fact* (real), the *magnitude* (oversold), and the *attribution* (unproven). All results come from a public toolkit (`quantlab/`) with a deterministic test suite.

2 Data and decomposition

For each trading day t with open/close auction prices, the close-to-close return splits *exactly* into overnight and intraday legs:

$$(1 + r_t^{\text{on}})(1 + r_t^{\text{id}}) = (1 + r_t^{\text{cc}}), \quad r_t^{\text{on}} = \frac{O_t}{C_{t-1}} - 1, \quad r_t^{\text{id}} = \frac{C_t}{O_t} - 1. \quad (1)$$

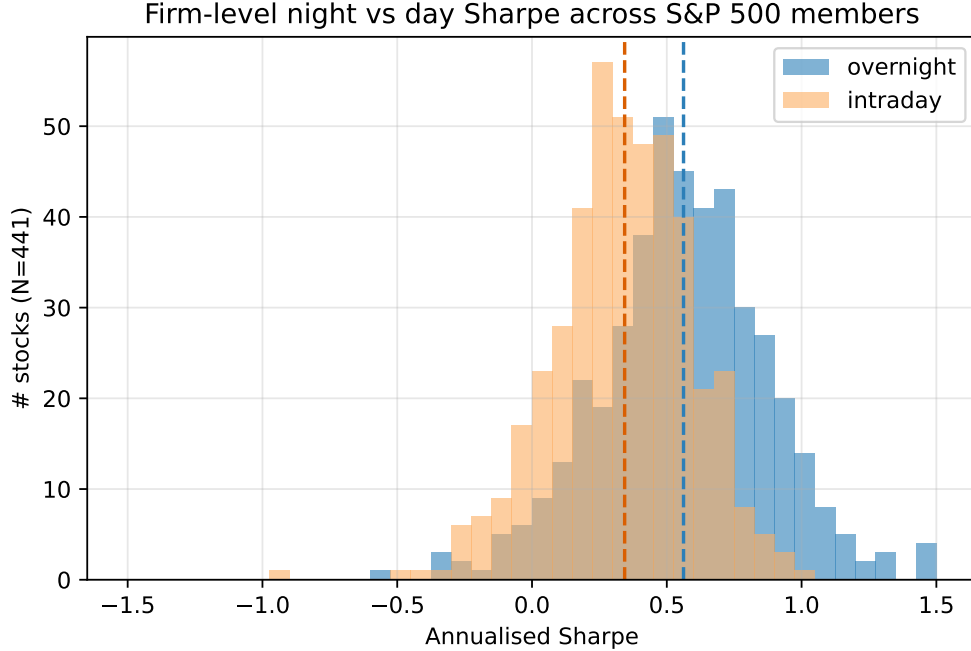


Figure 1: Firm-level overnight vs intraday annualised Sharpe across S&P 500 members. The overnight distribution is shifted right of the intraday one; dashed lines mark medians. The effect is a broad cross-sectional fact, not a few-index fluke. *Caveat:* current membership, hence survivorship bias.

There is no free parameter. Data are daily OHLC from Yahoo! Finance; the dividend-adjustment mode is treated as a modelling choice (ex-dividend occurs at the open), with `split_only` the default. The test suite verifies the identity to machine precision ($\sim 10^{-16}$).

3 The fact is real

On SPY the overnight mean is +3.25 bps/day with a Newey–West (HAC) $t \approx 4.9$ [11]; the intraday mean is insignificant ($t \approx 0.9$). The Lo [8] Sharpe-ratio t -statistic is ≈ 4.5 . The effect is not an artefact of assuming i.i.d. returns.

It is also not a handful of cherry-picked indices. At the *firm* level across S&P 500 members ($N \approx 441$ with clean post-2010 history), $\sim 69\%$ of stocks have a higher overnight than intraday Sharpe; the median overnight Sharpe (≈ 0.56) exceeds the median intraday Sharpe (≈ 0.34); and an equal-weight cross-section earns +372% overnight versus +233% intraday (HAC $t \approx 3.8$), echoing [9] (Figure 1). Finally, a White [13] Reality Check over a universe of major indices rejects the null that the best overnight Sharpe is a product of multiplicity ($p \approx 0$): on liquid indices the effect survives a data-snooping correction. We therefore concede the fact without reservation. The interesting questions are about magnitude, cause, and exploitability.

4 The magnitude is oversold

Three multipliers inflate the headline.

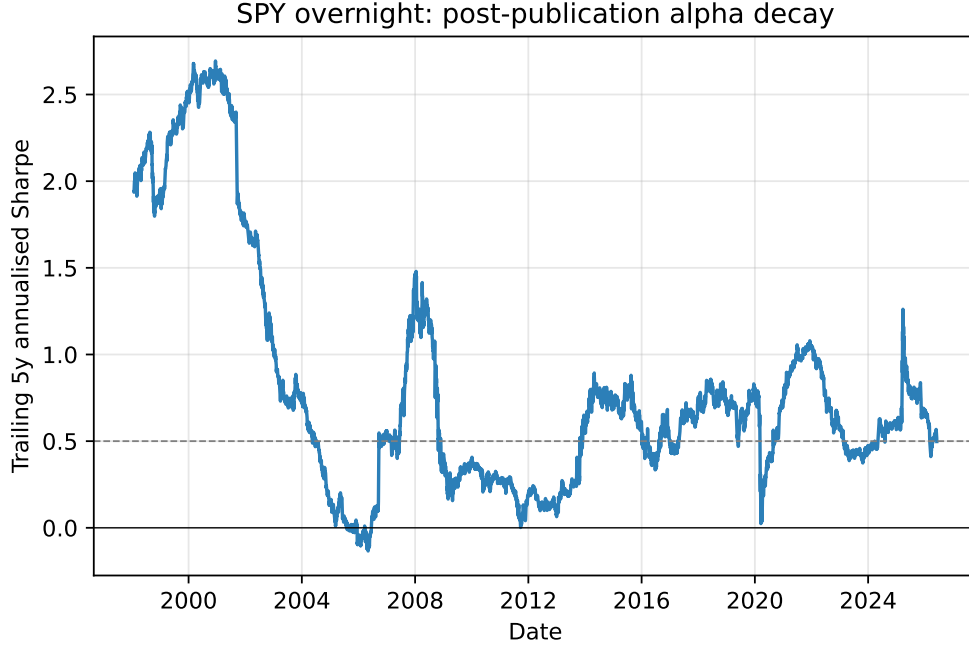


Figure 2: Trailing 5-year annualised Sharpe of the SPY overnight leg. An effect that fades as it becomes documented is the opposite of an ongoing, deliberate scheme.

Compounding on a log axis. A *constant, innocent* 1 bps/night drift compounds to +124% over 32 years; 30 bps/night to trillions of percent. The explosion is the exponent, not a mechanism.

The clock illusion. The overnight window is not 6.5 hours like the day: it runs from the 16:00 close to the 09:30 open (17.5h on weeknights) and across weekends and holidays (65h+), averaging ~ 28 calendar hours. Placing both legs on a per-hour footing collapses SPY’s overnight advantage from $\sim 4.1\times$ *per session* to $\sim 1.3\times$ *per calendar hour*. Most of the famous gap is a unit error; a modest genuine premium remains.

Data artefacts. A few mis-adjusted split/dividend prints mechanically shovel return from the day into the night; an automatic detector flags them. These drive the most extreme emerging-market figures, and selecting “the 25 most problematic” from a large universe guarantees significance by construction.

5 What survives is beta, untradable, and decaying

Holding the market every night is bearing gap risk continuously. Regressing the overnight leg on the market (close-to-close) gives $\beta \approx 0.33$: roughly 40% of SPY’s overnight return is a gap-risk premium, not distinct alpha. The residual alpha (≈ 1.9 bps/day) sits *below* the break-even round-trip cost (≈ 3.3 bps): even the non-beta part does not survive execution. A cost sweep turns a positive gross Sharpe negative at a realistic 5 bps round trip—precisely the fate of the NSPY/NIWM “night effect” ETFs (2022–2023). And the edge is fading: the trailing 5-year overnight Sharpe has fallen from ~ 2 (1998) to ~ 0.5 (2026), the textbook post-publication decay of [10] (Figure 2).

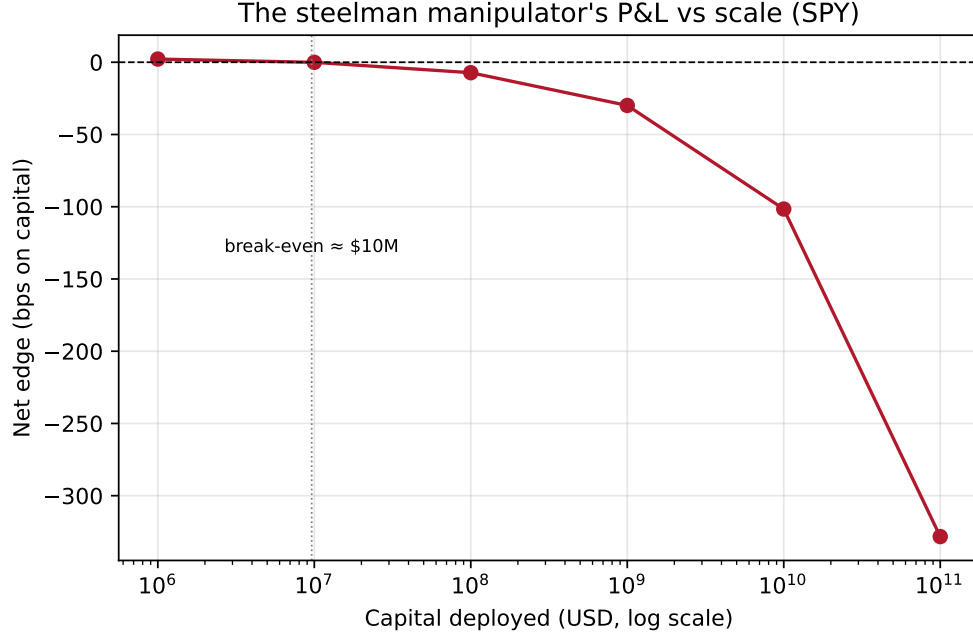


Figure 3: Net edge (bps on capital) of the alleged round-trip overnight harvester versus deployed capital. Profitable only at trivial scale; deeply negative at the sizes a “world-moving” actor would require.

6 Capacity and the steelman manipulator

Even a real, free edge is worthless if it cannot hold capital. Under the square-root market-impact law [1], round-trip impact equals SPY’s gross overnight edge at only $\sim \$10\text{M}$ of capital. We then *steelman* the manipulation thesis: grant a firm the *entire* overnight drift on its capital and charge it only round-trip temporary impact (permanent impact nets to zero over a symmetric round trip, the no-dynamic-arbitrage condition of [5, 4]). Net profit is

$$\Pi(K) = K \left(r^{\text{on}} - 2c\sigma\sqrt{K/\text{ADV}\$} \right), \quad (2)$$

positive only for small K . The break-even is $\sim \$10\text{M}$; at $\$1\text{B}$ the firm loses $\sim \$0.75\text{B}/\text{year}$, at $\$10\text{B} \sim \$25\text{B}/\text{year}$ (Figure 3). A firm large enough to move world markets could not profit from an edge this thin: the mechanism is self-defeating at the scale the accusation requires.

7 Is it manipulation? A Bayesian reading

Write the posterior odds for the manipulation hypothesis M against “risk premium + microstructure” S given evidence E_i :

$$\frac{P(M | E)}{P(S | E)} = \frac{P(M)}{P(S)} \prod_i \frac{P(E_i | M)}{P(E_i | S)}. \quad (3)$$

The basic night > day pattern is near-equally likely under both hypotheses, so it barely discriminates ($\text{LR} \approx 1$). The *discriminating* evidence—foreign-ETF sign inversions driven by the listing clock, the Chinese T+1 inversion [12], the clock illusion, the negative net P&L at scale, and the post-publication decay—all favour S . With deliberately moderate, stated likelihood ratios and a prior of 0.5, the posterior probability of manipulation is $\approx 2.4\%$, and it remains low across a wide

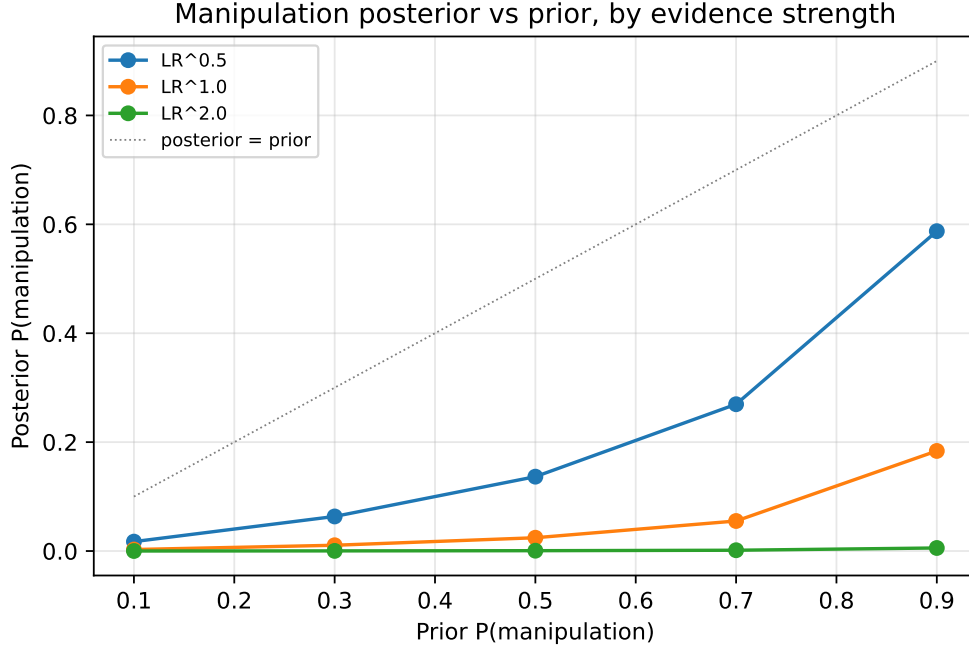


Figure 4: Posterior probability of manipulation versus prior, for three evidence strengths (LR raised to powers 0.5/1/2). The diagonal is “posterior = prior”. Discriminating evidence pulls every curve below the diagonal.

range of priors and evidence strengths (Figure 4); only a strong prior combined with deliberately weakened evidence keeps it elevated. The likelihood ratios are subjective and combined under a generous independence assumption—stated openly, not hidden—so the robust conclusion is directional: the evidence moves a reasonable observer *toward* microstructure, not toward fraud.

8 Conclusion

The overnight anomaly is real, broad, and genuinely interesting—and largely *explained* by ordinary risk premia and market microstructure. Its magnitude is inflated by compounding and a calendar-time confound; what remains is mostly beta, is below trading costs, has decayed, and cannot scale. The manipulation attribution is not supported: the pattern is weak evidence, the discriminating evidence points the other way, and the alleged mechanism is economically self-defeating. A beautiful feature of microstructure was dressed up as both a free lunch and a global fraud; the honest, reproducible answer is more interesting than either.

Reproducibility. All figures, tables and numbers are produced by the open-source toolkit at the URL on the title page: a deterministic, tested Python package with continuous integration, executed notebooks, and this paper’s figure script (`paper/make_figures.py`). *Not investment advice.*

References

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